

Mean value of a function

Basic skills

Q1. a) $\int \sqrt[5]{x^4} dx = \int x^{\frac{4}{5}} dx = \frac{5}{9} x^{\frac{9}{5}} (+c)$

b) $\int e^x - \frac{1}{x} dx = e^x - \ln x$

c) $\int \sin x - (3x+1)^5 dx = -\cos x - \frac{1}{18} (3x+1)^6$

Q2. $\int_0^9 \sqrt{x} dx = \int_0^9 x^{\frac{1}{2}} dx = \left[\frac{2}{3} x^{\frac{3}{2}} \right]_0^9$
 $= \left[\frac{2}{3} (\sqrt{x})^3 \right]_0^9 = \left(\frac{2}{3} \times \sqrt{9}^3 - 0 \right) = 18$

Q3. $f(x) = x^2$

a) $\int_1^5 x^2 dx = \left[\frac{1}{3} x^3 \right]_1^5 = \frac{125}{3} - \frac{1}{3} = \frac{124}{3}$

b) For average 4 units so $\frac{1}{4} \times \frac{124}{3} = \frac{31}{3}$

Q4. $\frac{1}{16-4} \int_4^{16} \sqrt{x} \, dx = \frac{1}{12} \int_4^{16} x^{\frac{1}{2}} \, dx$

$$= \frac{1}{12} \left[\frac{2}{3} x^{\frac{3}{2}} \right]_4^{16} = \frac{1}{12} \left[\frac{2}{3} (\sqrt{x})^3 \right]_4^{16}$$

$$= \frac{1}{12} \left(\frac{2}{3} \sqrt{16}^3 - \frac{2}{3} \sqrt{4}^3 \right) = \frac{1}{12} \left(\frac{128}{3} - \frac{16}{3} \right)$$

$$= \frac{1}{12} \times \frac{112}{3} = \frac{28}{9}$$

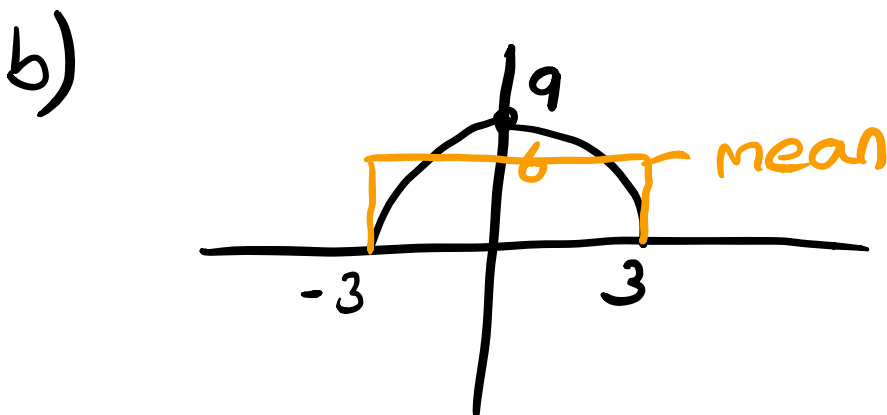
Competency skills

Q5. $y = 9 - x^2 \quad [-3, 3]$

a) $\frac{1}{3 - -3} \int_{-3}^3 9 - x^2 \, dx = \frac{1}{6} \left[9x - \frac{1}{3} x^3 \right]_{-3}^3$

$$= \frac{1}{6} \left((27 - 9) - (-27 + 9) \right) = \frac{1}{6} (18 - -18)$$

$$= \frac{1}{6} \times 36 = 6$$



Q6. a) $\frac{1}{2-0} \int_0^2 e^x dx = \frac{1}{2} [e^x]_0^2$
 $= \frac{1}{2} (e^2 - e^0) = \frac{e^2 - 1}{2}$

b) $\frac{1}{4-1} \int_1^4 \frac{1}{\sqrt{x}} dx = \frac{1}{3} \int_1^4 x^{-\frac{1}{2}} dx = \frac{1}{3} [2x^{\frac{1}{2}}]_1^4$
 $= \frac{1}{3} [2\sqrt{x}]_1^4 = \frac{1}{3} (2\sqrt{4} - 2\sqrt{1}) = \frac{1}{3} (4 - 2)$
 $= \frac{2}{3}$

c) $\frac{1}{2-1} \int_1^2 \frac{1}{x} dx = \frac{1}{1} [\ln x]_1^2 = \ln 2 - \ln 1$
 $= \ln 2$

Q7. $\frac{1}{2-0} \int_0^2 (2x-4)^4 dx = \frac{1}{2} \left[\frac{1}{10} (2x-4)^5 \right]_0^2$
 $= \frac{1}{2} \left(0 - \frac{1}{10} (-4)^5 \right) = \frac{1}{20} 4^5 = \frac{256}{5}$

$\frac{1}{4-0} \int_0^4 (2x-4)^4 dx = \frac{1}{4} \left[\frac{1}{10} (2x-4)^5 \right]_0^4$
 $= \frac{1}{4} \left(\frac{1}{10} 4^5 - \frac{1}{10} (-4)^5 \right) = \frac{1}{4} \times \frac{2}{10} 4^5 = \frac{1}{20} 4^5 = \frac{256}{5}$

So same mean

Advanced skills

Q8. $y = ax^2$ $[0, b]$, $y = bx^2$ $[0, a]$

$$\frac{1}{b} \int_0^b ax^2 dx = \frac{a}{b} \left[\frac{1}{3} x^3 \right]_0^b = \frac{a}{b} \left(\frac{1}{3} b^3 - 0 \right)$$
$$= \frac{ab^2}{3}$$

$$\frac{1}{a} \int_0^a bx^2 dx = \frac{b}{a} \left[\frac{1}{3} x^3 \right]_0^a = \frac{b}{a} \left(\frac{1}{3} a^3 - 0 \right)$$
$$= \frac{a^2 b}{3}$$

so need $\frac{a^2 b}{3} = \frac{ab^2}{3}$

$$\Rightarrow a^2 b - ab^2 = 0 \quad ab(a-b) = 0$$

so either $a=0$ or $b=0$ or $a=b$

Q9. $f(x) = \sin x$

a) $\frac{1}{2\pi} \int_0^{2\pi} \sin x dx = \frac{1}{2\pi} [-\cos x]_0^{2\pi} = \frac{1}{2\pi} (-1 - -1)$
 $= 0$ Area is zero

b) $\sqrt{\frac{1}{2\pi} \int_0^{2\pi} \sin^2 x dx} = \sqrt{\frac{1}{2\pi} \int_0^{2\pi} \frac{1}{2} - \frac{1}{2} \cos 2x dx}$
↗
 $\cos 2x = 1 - 2\sin^2 x$

$$\sqrt{\frac{1}{2\pi} \left[\frac{1}{2}x - \frac{1}{4}\sin 2x \right]_0^{2\pi}} = \sqrt{\frac{1}{2\pi} (\pi - 0) - (0 - 0)}$$

$$= \sqrt{\frac{1}{2\pi} \pi} = \sqrt{\frac{1}{2}} = \frac{1}{\sqrt{2}}$$

c) The root mean square helps by making values positive for area (counteracted by square root)

Q10. $\frac{1}{4} \int_0^4 f(x) dx = 2 \Rightarrow \int_0^4 f(x) dx = 8$

$\frac{1}{2} \int_4^6 f(x) dx = 10 \Rightarrow \int_4^6 f(x) dx = 20$

so $\int_0^6 f(x) dx = \int_0^4 f(x) dx + \int_4^6 f(x) dx$

$$= 8 + 20 = 28$$

so mean over $[0, 6]$ is

$$\frac{1}{6} \int_0^6 f(x) dx = \frac{1}{6} \times 28 = \frac{14}{3}$$